

MSA GRR Study Guide - % Study Variation vs % Tolerance

Understanding both GRR acceptance criteria | AIAG MSA 4th Edition

QUALITY HEAD AGENTS | AIAG MSA 4th Edition | IATF 16949:2016

What is a GRR Study?

- GRR (Gauge Repeatability and Reproducibility) measures the total variation in your measurement system.
- $GRR = \sqrt{EV^2 + AV^2}$ where EV = Equipment Variation and AV = Appraiser Variation
- The GRR must be small relative to both: (1) Total Study Variation (TV) and (2) Tolerance (USL-LSL)
- AIAG standard: 10 parts x 2-3 operators x 2-3 trials = 40-90 measurements minimum

% GRR / Study Variation (Process Control Focus)

Formula:	$\% GRR/TV = (GRR / TV) \times 100$, where $TV = \sqrt{GRR^2 + PV^2}$
< 10%:	Acceptable - Gauge is suitable for process control and SPC
10% - 30%:	Conditional - May be acceptable based on importance of application, gauge cost
> 30%:	Unacceptable - Gauge must be improved or replaced before use
Key Use:	Use % Study Var when controlling ongoing process variation with SPC

% GRR / Tolerance (Inspection Decision Focus)

Formula:	$\% GRR/Tol = (5.15 \times \sigma_{GRR} / Tolerance) \times 100$
< 10%:	Acceptable - Gauge is suitable for inspection decisions
10% - 30%:	Conditional - Increasing misclassification risk
> 30%:	Unacceptable - Gauge will produce excessive false accepts and false rejects
Key Use:	Use % Tolerance when making accept/reject decisions at inspection

Which Criterion Matters More?

- Both criteria must be evaluated and reported in PPAP Element 8.
- A gauge can pass % Study Var but fail % Tolerance - both must be reviewed.
- For CC (Critical Characteristics): target < 10% on BOTH criteria.
- For SC (Significant Characteristics): < 30% on both is minimum.
- Auditor will ask: 'Show me GRR vs tolerance for your CC characteristics.'

NDC - Number of Distinct Categories

Formula:	$NDC = 1.41 \times (PV / GRR)$, rounded down
NDC >= 5:	Acceptable - Gauge can distinguish at least 5 different levels
NDC = 2-4:	Marginal - Gauge can only crudely classify product
NDC = 1:	Unacceptable - Gauge cannot distinguish between parts at all
IATF Audit:	Auditor will verify NDC is reported and >= 5 for variable gauges